ATLAS Injection Eval-Only Metrics

Detailed description of the diagnostics computed by `atlas_injection_eval_only.py` over cached spoofed predictions and a clean full-validation result.pkl.

Setup

For each attacked (target) frame, five IoU matrices are precomputed once and reused by all metric stages:

- IoU(all attacked preds, spoof GT) — for preserved ASR
- IoU(attacked vehicle preds, spoof GT) — for score-thresholded ASR, IoU distribution, FP counts
- IoU(attacked vehicle preds, real GT vehicles) — for FP exclusion + displacement matching
- IoU(clean vehicle preds, spoof GT) — for clean-baseline FP counts
- IoU(clean vehicle preds, real GT vehicles) — for displacement matching + clean FP exclusion

All IoU is exact rotated 3D IoU: BEV Sutherland-Hodgman polygon clipping × z-axis height overlap. Same math as pcdet's `boxes_iou3d_gpu`, computed on CPU with a numba-compiled parallel kernel.

1. Preserved ASR (original semantics, unchanged)

For each target frame:

- Compute IoU3D between the spoof GT and every prediction (all classes).
- Take the argmax prediction and inspect its class label:
  - — Vehicle (label 1): counts as FP if IoU ≥ 0.7
  - — Pedestrian (label 2) or Cyclist (label 3): FP if IoU ≥ 0.5
- Frames with zero predictions are counted separately.

Reported: ASR (vehicle FP rate), per-class ASR, vehicle-FP count, zero-prediction rate, confidence stats of vehicle FPs, best-IoU distribution.

2. Score-thresholded ASR

For each score threshold $s \in \{0.1, 0.3, 0.5, 0.7\}$ and each target frame:

- Filter to vehicle preds with score ≥ s.
- Compute max IoU3D between spoof GT and those preds (0 if none).
- Frame contributes to ASR(s) if that max IoU ≥ asr_iou (0.7).

Reported: ASR(s) for each threshold. Shows how ASR degrades as you raise the confidence bar.

3. IoU distribution

Same per-frame max IoU values as (2), but reported as a distribution rather than thresholded.

Reported per score gate: min, q1, median, mean, q3, max, and fraction ≥ 0.7 (which recovers standard ASR). Primary block uses the lowest score gate (0.1) — shows whether “misses” are near-misses or total whiffs.

4. Box displacement (attraction toward spoof)

For each target frame and each real GT vehicle:

- Find the clean prediction best matching that GT — max IoU among vehicle preds with score ≥ 0.1, requiring IoU ≥ match_iou (0.7). If none, skip (real object not detected cleanly, unmeasurable).
- Find the attacked prediction matching the same GT by the same criterion.
- If clean matched but attacked didn't → count as **disappearance** (not forced into a displacement value).
- If both matched, compute:
  - — $\Delta c = c_{attacked} - c_{clean}$ — center shift (3D)
  - — $sep = c_{spoof} - c_{clean}$ — clean→spoof vector
  - — $d_{toward} = \Delta c \cdot unit(sep)$ — component of shift along clean→spoof (meters, signed)
  - — $\rho = d_{toward} / ||sep||$ — normalized: 0 = didn't move toward, 1 = moved all the way to spoof

“Primary target” per frame = the real GT vehicle whose clean prediction is nearest the spoof (the one most plausibly being dragged). Reported separately.

Reported: count of clean-matched vehicles, matched pairs, disappearances, fraction with $d_{toward} > 0$, and full stats (min/q1/median/mean/q3/max) for d_{toward}, ρ, displacement magnitude $||\Delta c||$, and clean→spoof separation.

5. Spoof-region false positives

Crossed over $\tau_{spoof} \in \{0.3, 0.5, 0.7\}$ × score $s \in \{0.1, 0.3, 0.5, 0.7\}$. For each ($\tau$, s) and each target frame, count vehicle preds satisfying all three:

- score ≥ s
- IoU3D(spoof GT) ≥ τ
- max IoU3D across all real GT vehicles < match_iou (0.7) — i.e., the pred isn't matching a legitimate object

Repeat on the clean prediction for the same frame → **clean baseline count**.

Attack-induced count — two variants reported:

- *Clean-gated rule:* if any qualifying phantom already exists in the clean frame, don't credit the attack for that frame (attack_induced = 0 if clean_count > 0 else attack_count).
- *Subtractive rule:* max(0, attack_count – clean_count).

Reported per (τ, s): total FPs, per-frame mean, phantom detection rate (fraction of frames with ≥1 FP), per-frame count distribution stats, FP confidence stats, FP IoU-with-spoof stats, clean baseline totals + rate, both attack-induced variants.

Output Artifacts

Written atomically to `--out_dir`:

- `detailed_metrics.md` — human-readable summary
- `detailed_metrics.json` — same data, without large raw arrays
- `detailed_summary.pkl` — full data including raw per-frame arrays and per-record displacement entries

The `meta` block records: detector, variant, thresholds, num_segments, seed, selected segment list (for reproducibility), and input paths.